

Capability: Build your own proprietary learning environments within the tenant boundary to train or tune models, where models learn against real workflows without exposing the company’s knowledge.

Choice: Ensure the orchestration layer is decoupled from any single model. Ask yourself: If any one model you are using is taken away, do you still have the ability to operate and optimize for your evals using other models? Does your company “veteran” capability remain with you even if a given “generalist” model is taken away?

Cost: By decoupling the orchestration layer, you are also able to bring together context, models, and tasks in the most efficient and cost-effective way without sacrificing quality.

Compound: Bring these four together and you create your own continuous learning loop (i.e. hill climbing machine) that will allow your AI investments to compound the value of your firm.

In other words, a company should be able to use a model without giving up the knowledge that makes it unique. That is the reverse information paradox we need to confront.

GPT-6.1 Sol Ultrafast in Codex and ChatGPT Work

CODEX CHANGELOG · 08 OCT 2026 · [SOURCE](#)

GPT-6.1 Sol Ultrafast in Codex and ChatGPT Work

Ultrafast mode speeds up token generation with GPT-6.1 Sol in Codex and ChatGPT Work.

Ultrafast is available on Pro \$500 and eligible Enterprise and Edu plans. Enterprise access is off by default; workspace owners must enable it. GPT-6.1 Sol Ultrafast supports inference residency in the United States and Europe (EEA + Switzerland).

See [Ultrafast mode](/codex/agent-configuration/speed#ultrafast-mode) for plan and workspace requirements, [Models](/codex/models#gpt-61-sol) for model guidance, and [Pricing](/codex/pricing) for usage and credit rates.

FRED TALKS

9th October 2026

CONTENTS

Big tech engineers need big egos

SEANGOEDECKE.COM RSS FEED · 1975 WORDS

Control the ideas, not the code

ANTIREZ.COM · 1245 WORDS

Search has its own bitter lesson

DOUG TURNBULL · 1012 WORDS

Your Agent is a Distributed System (and fails like one)

MAHESH'S BLOG · 1053 WORDS

The new rules of context engineering for Claude 5 generation models

CLAUDE.COM · 1148 WORDS

The Reverse Information Paradox

SN SCRATCHPAD · 819 WORDS

GPT-6.1 Sol Ultrafast in Codex and ChatGPT Work

CODEX CHANGELOG · 89 WORDS

Big tech engineers need big egos

SEANGOEDECKE.COM RSS FEED · 14 MAR 2026 · [SOURCE](#)

It's a common position among software engineers that big egos have no place in tech¹. This is understandable - we've all worked with some insufferably overconfident engineers who needed their egos checked - but I don't think it's correct. In fact, I don't know if it's possible to survive as a software engineer in a large tech company without some kind of big ego.

However, it's more complicated than "big egos make good engineers". The most effective engineers I've worked with are simultaneously high-ego in some situations and surprisingly low-ego in others. What's going on there?

Engineers need ego to work in large codebases

Software engineering is shockingly humbling, even for experienced engineers. There's a reason this joke is so popular:



The minute-to-minute experience of working as a software engineer is dominated by *not knowing things* and *getting things wrong*. Every time you sit down and write a piece of code, it will have several things wrong with it: some silly things, like missing semicolons, and often some major things, like bugs in the core logic. We spend most of our time fixing our own stupid mistakes.

On top of that, even when we've been working on a system for years, we still don't know that much about it. I wrote about this at length in *Nobody knows how large software products work*, but the reason is that big codebases are just that complicated. You simply can't confidently answer questions about them without going and doing some research, even if you're the one who wrote the code.

This requires more than data protection. Models learn from "exhaust," the prompts people write, the tools agents use, and especially the corrections people make when the model is wrong. Every correction is distilled into institutional know-how. It's the kind of knowledge a competitor could never buy, and the kind that leaks almost imperceptibly: trace by trace, correction by correction, eval by eval.

In consuming intelligence, you are creating intelligence. And what you create should belong to you. This is your particular intelligence, in Hayek's sense: the knowledge of time, place, and circumstance that no one else can hold. It knows what you think, what you value, and how you measure success.

While the great innovation that comes from model providers having fair use rights to train models on public data is needed, I find it ironic that the status quo is to then turn around and impose restrictive terms on distillation, and to reserve the right to learn from customer usage and interaction data. If learning flows in only one direction, economic value converges toward the owners of the learning infrastructure rather than the creators of the knowledge itself. Therefore, it's imperative that we distribute the learning infrastructure to every firm so that they can control their own learning loop.

As Alex Karp put it: "What the technical customers want is control over their compute, their models, their data stack, and their alpha. They want to know they own the means of production, and it's not being transferred to someone else." The current regime does precisely the transfer Karp and companies fear.

That is why enterprises need a real trust boundary for their human capital and token capital to compound. It is where an organization's data, traces, evals, adapted weights, and memory accumulate and improve together. And it is a hard boundary across which nothing crosses, not even the intelligence exhaust, without consent. Enterprises will demand the rights to use model outputs to fine tune and/or train their own models. I think of this as every firm's right to align models to their enterprise accountability obligations.

In the cloud era, enterprises accumulated data. In the AI era, they accumulate learning. The trust boundary must evolve accordingly, from protecting information to protecting the mechanisms through which organizations learn, adapt, and compound intelligence. There are a few things every enterprise must do to ensure this:

Control: Create your private evals, because evals define what "good" looks like inside the organization. Also, retain ownership of your organization's memory, traces, feedbacks, decisions, and institutional context, and ability to use outputs of models from your own tasks and queries.

Try simplifying

Across your system prompt, skills, and CLAUDE.md files, you may need to simplify just like we did. We rolled out a new command called ``claude doctor`` which will help you do this automatically as well. For more details on prompting more advanced models specifically, check out our [Fable field guide](#).

This article was written by Thariq Shihpar, member of technical staff, Anthropic.

The Reverse Information Paradox

SN SCRATCHPAD · 05 AUG 2026 · [SOURCE](#)

Notes on advances in technology and real-world impact

In the age of intelligence, how should firms protect their core IP?

Nobel Prize winning economist Kenneth Arrow [famously described](#) a paradox in the market for information. “Its value for the purchaser is not known until he has the information, but then he has in effect acquired it without cost.” In Arrow’s “Information Paradox,” the seller risks giving away knowledge in order to sell it.

AI creates the reverse problem. In the AI age, the buyer risks giving away knowledge, just in order to use what they bought.

You essentially pay for intelligence twice, once with money, and again with something even more valuable: the proprietary knowledge you must reveal to make that intelligence useful. The better you want the model to perform, the more of that knowledge you have to feed it!

Over time, the information asymmetry becomes increasingly skewed. The seller learns more and more about you as you use what you purchased, while you learn very little about what the seller is learning in return.

That is what I think of as the Reverse Information Paradox.

Patents solve one aspect of Arrow’s paradox. They let an inventor disclose an idea without simply giving it away. The Reverse Information Paradox needs its own equivalent.

When you have to build something new or fix a tricky problem, it can often feel straight-up impossible to begin, because good software engineers know just how ignorant they are and just how complex the system is. You just have to throw yourself into the blank sea of millions of lines of code and start wildly casting around to try and get your bearings.

Software engineers need the kind of ego that can stand up to this environment. In particular, they need to have a firm belief that they *can* figure it out, no matter how opaque the problem seems; that if they just keep trying, they can break through to the pleasant (though always temporary) state of affairs where they understand the system and can see at a glance how bugs can be fixed and new features added².

Engineers need ego to work in big tech companies

What about the non-technical aspects of the job? Nobody likes working with a big ego, right? Wrong. Every great software engineer I’ve worked with in big tech companies has had a big ego - though as I’ll say below, in some ways these engineers were surprisingly low-ego.

You need a big ego to take positions. Engineers love being non-committal about technical questions, because they’re so hard to answer and there’s often a plausible case for either side. However, as I [keep saying](#), engineers have a duty to take clear positions on unclear technical topics, because the alternative is a non-technical decision maker (who knows even less) just taking their best guess. It’s scary to make an educated guess! You know exactly all the reasons you might be wrong. But you have to do it anyway, and ego helps *a lot* with that.

You need a big ego to be willing to make enemies. Getting things done in a large organization means making some people angry. Of course, if you’re making lots of people angry, you’re probably screwing up: being too confrontational or making obviously bad decisions. But if you’re making a large change and one or two people are angry, that’s just life. In big tech companies, any big technical decision will affect a few hundred engineers, and one of them is bound to be unhappy about it. You can’t be so conflict-averse that you let that stop you from doing it, if you believe it’s the right decision. In other words, you have to have the confidence to believe that you’re right and they’re wrong, even though technical decisions always involve unclear tradeoffs and it’s impossible to get absolute certainty.

You need a big ego to correct incorrect or unclear claims. When I was still in the philosophy world, the Australian logician Graham Priest had a reputation for putting his hand up and stopping presentations when he didn’t understand something that was said, and only allowing the seminar to continue when he felt like he understood. From his perspective, this wasn’t rude: after all, if *he* couldn’t understand it, the rest of the audience probably couldn’t either, and so he was doing them a favor by forcing a more clear explanation from the speaker.

This is obviously a sign of a big ego. It's also a trait that you need in a large tech company. People often nod and smile their way past incorrect technical claims, even when they suspect they might be wrong - assuming that they've just misunderstood and that somebody else will correct it, if it's truly wrong. **If you are the most senior engineer in the room, correcting these claims is your job.**

If everyone in the room is so pro-social and low-ego that they go along to get along, decisions will get made based on flatly incorrect technical assumptions, projects will get funded that are impossible to complete, and engineers will burn weeks or months of their careers vainly trying to make these projects work. You have to have a big enough ego to think "actually, I think I'm right and everyone in this room is confused", even when the room is full of directors and VPs.

Sometimes you need to put your ego aside

All of this selects for some pretty high-ego engineers. But in order to actually *succeed* in these roles in large tech companies, you need to have a surprisingly low ego at times. **I think this is why *really* effective big tech engineers are so rare: because it requires such a delicate balance between confidence and diffidence.**

To be an effective engineer, you need to have a towering confidence in your own ability to solve problems and make decisions, even when people disagree. But you also need to be willing to instantly subordinate your ego to the organization, when it asks you to. At the end of the day, your job - the reason the company pays you - is to execute on your boss's and your boss's boss's plans, whether you agree with them or not.

Competent software engineers are allowed quite a lot of leeway about *how* to implement those plans. However, they're allowed almost no leeway at all about the plans themselves. In my experience, being confused about this is a common cause of burnout³. Many software engineers are used to making bold decisions on technical topics and being rewarded for it. Those software engineers then make a bold decision that disagrees with the VP of their organization, get immediately and brutally punished for it, and are confused and hurt.

In fact, **sometimes you just get punished and there's nothing you can do.** This is an unfortunate fact of how large organizations function: even if you do great technical work and build something really useful, you can fall afoul of a political battle fought three levels above your head, and come away with a *worse* reputation for it. Nothing to be done! This can be a hard pill to swallow for the high-ego engineers that tend to lead really useful technical projects.

Applying this to your context

Pulling this all together, what does this look like when you assemble your context?

System Prompt

A system prompt is heavily tied to the product context. It tells Claude what product it's operating in and what it's doing. For Claude Code, you will likely never modify this, but if you are building your own agent harness, this is where you should spend a lot of time.

CLAUDE.md

Keep your CLAUDE.md lightweight and briefly describe what your repo is for, but spend most of the tokens on gotchas inside of the codebase. For example, you may organize your code to keep types in one monolithic file and nowhere else. Avoid stating 'the obvious' things Claude should know by looking at your file system or your repo.

Use progressive disclosure heavily, for example if you have several unique instructions on how to verify your work, create a verification skill and reference it from your CLAUDE.md.

Skills

Think of skills as lightweight guides to let Claude find information when needed. Avoid making them overconstrained, except in highly important areas.

For long skills, try and use progressive disclosure as much as possible- divide it into many files and split them out.

It's best when skills encode particular opinions, knowledge, or best practices that are particular to you, your team, or product.

References

You can @ mention files to include them as references. References allow Claude to refer to in-depth information about the current plan.

This might be in specs files, mockups, or even entire codebases. Generally you should prefer files that are in code as it provides clear, high-fidelity instructions to Claude in a language it knows very well. For example, a HTML mockup of a design will generally produce better results than a description of the design or a screenshot.

Then: Repeat yourself

Now: Simple tool descriptions

Earlier Claude models could sometimes need repeated instructions or be more likely to listen to instructions at the end of their context window than at the start. This meant our system prompt would sometimes have references to tools in the main system prompt as well as instructions in the tool description.

We found we could delete these repeat examples and put instructions on how to use tools in the tool descriptions rather than the system prompt.

Then: Memory in CLAUDE.md files

Now: Auto-memory

We used to encourage users to save things to Claude’s memory, by using the # hotkey to write to their CLAUDE.md automatically. Instead, Claude now automatically saves memories that are relevant to the work and to you.

Then: Simple specs

Now: Rich references

In plan mode, Claude Code has heavily relied on markdown files with plans. Storing these files as plans helped Claude refer to them when needed. Another similar best practice was to store specs in the codebase for Claude to refer to while working across longer projects.

But we’ve found that Claude can handle increasingly more complicated references. Instead of simple markdown files, Claude can reference HTML artifacts created by our new artifacts feature.

You may also give Claude references in the form of code. A spec may also be a detailed test suite, or a function in a different codebase that Claude might port.

Rubrics are another form of references. Rubrics allow Claude to try and verify your taste in a particular field (e.g. what does a good API design look like) by using dynamic workflows and spinning up verifier agents with those rubrics.

You also have to be okay with having your projects cancelled at the last minute. It’s a very common experience in large tech companies that you’re asked to deliver something quickly, you buckle down and get it done, and then right before shipping you’re told “actually, let’s cancel that, we decided not to do it”. This is partly because the decision-making process can be pretty fluid, and partly because many of these asks originate from off-hand comments: the CTO implies that something might be nice in a meeting, the VPs and directors hustle to get it done quickly, and then in the next meeting it becomes clear that the CTO doesn’t actually care, so the project is unceremoniously cancelled⁴.

Final thoughts

Nobody likes to work with a bully, or with someone who refuses to admit when they’re wrong, or with somebody incapable of empathy. But you really do need a strong ego to be an effective software engineer, because software engineering requires you to spend most of your day in a position of uncertainty or confusion. If your ego isn’t strong enough to stand up to that - if you don’t believe you’re good enough to power through - you simply can’t do the job.

This is particularly true when it comes to working in a large software company. Many of the tasks you’re required to do (particularly if you’re a senior or staff engineer) require a healthy ego. However, there’s a kind of catch-22 here. If it insults your pride to work on silly projects, or to occasionally “catch a stray bullet” in the organization’s political fights, or to have to shelve a project that you worked hard on and is ready to ship, you’re too high-ego to be an effective software engineer. But if you can’t take firm positions, or if you’re too afraid to make enemies, or you’re unwilling to speak up and correct people, you’re too low-ego.

Engineers who are low-ego in general can’t get stuff done, while engineers who are high-ego in general get slapped down by the executives who wield real organizational power. The most successful kind of software engineer is therefore a chameleon: low-ego when dealing with executives, but high-ego when dealing with the rest of the organization⁵.

1. What do I mean by “ego”, in this context? More or less the colloquial sense of the term: a somewhat irrational self-confidence, a tendency to believe that you’re very important, the sense that you’re the “main character”, that sort of thing

↩

2. Why is this “ego”, and not just normal confidence? Well, because of just how murky and baffling software problems feel when you start working on them. You really do need a degree of confidence in yourself that feels unreasonable from the inside. It should be

obvious, but I want to explicitly note that you don't *just* need ego: you also have to be technically strong enough to actually succeed when your ego powers you through the initial period of self-doubt.

↵

3. I share the increasingly-common view that burnout is not caused by working too hard, but by hard work unrewarded. That explains why nothing burns you out as hard as being punished for hard work that you expected a reward for.

↵

4. It's more or less exactly [this scene](#) from Silicon Valley.

↵

5. This description sounds a bit sociopathic to me. But, on reflection, it's fairly unsurprising that competent sociopaths do well in large organizations. Whether that kind of behavior is worth emulating or worth avoiding is up to you, I suppose.

↵

Control the ideas, not the code

ANTIREZ.COM · 18 JUL 2026 · [SOURCE](#)

Look at the past history of this blog. There are many blog posts about

So mine is a trick. People feel more and more programming is complete

This is why yesterday, on X, I said that I believe many programmers are

1. You can now generate a lot of code, even **not** accounting for the
2. LLMs are very good at writing locally optimal code, and are worse
3. The working day is 8 hours. If you read the code, it is a tradeoff

Then: Give Claude examples

Now: Design interfaces

The number one rule for tool usage was to give Claude examples on how to use them. With our newest models, we've found that giving examples actually constrains them to a certain exploration space.

Instead of using examples, think more about the design of your tools, scripts and files- what parameters does Claude have and how can they be more expressive?

For example, in the Todo tool example, just listing status as an enumeration between pending, in_progress, and completed, hints to Claude about how to use it. The instruction on keeping one item in_in_progress helps define our requested behavior.

Then: Put it all upfront

Now: Use progressive disclosure

Because Claude Code was focused on coding, our system prompt included detailed information on how to do code review and verification. These were not always needed, but when they were, it was crucial information.

Since then, Claude Code has gotten very competent at using progressive disclosure- loading the right context at the right times. For example, we moved verification and code review into their own skills that Claude Code could selectively call.

But progressive disclosure is not just for skills, we also use it for tools. Some of our tools are 'deferred loading,' which means the agent must search for their full definitions using ToolSearch before using them. This allows us to have more tools (such as our Task tools) that don't take up context until they're needed.

The same can be applied to your own CLAUDE.md and Skill.md files. A common myth is that you want to make these a central repository for every known practice that you *might* run into, because Claude would not find it otherwise. Instead, [consider having a tree of files that can be loaded at the right time](#).

But this is just one step. A self-writing program is a strange, astonishing, and novel creature; making it reliable will require innovation in every branch of Computer Science!

The new rules of context engineering for Claude 5 generation models

CLAUDE.COM · 09 AUG 2026 · [SOURCE](#)

Then and now

There were a number of previous context engineering best practices that had become myths. Including:

Then: Give Claude rules

Now: Let Claude use judgement

When we first rolled out Claude Code, we needed to be sure that Claude avoided worst case scenarios, such as deleting files. This meant we would give particularly strong guidance that might not always be true, For example, in the system prompt we used to say:

In code: default to writing no comments. Never write multi-paragraph docstrings or multi-line comment blocks — one short line max. Don't create planning, decision, or analysis documents unless the user asks for them — work from conversation context, not intermediate files.

But for a certain subset of prompts, this guidance would be wrong. In the case of documentation, the user may have their own preferences, or specific parts of very complex code might need multi-line comment blocks.

Still, without these guardrails for older models, the comments Claude wrote would be incorrect in many cases and we had to accept this tradeoff. But newer models have better judgement and can handle these decisions well without explicit rules.

In the new system prompt we say: *Write code that reads like the surrounding code: match its comment density, naming, and idiom.*

Controlling the ideas. Do you remember this phrasing from the Mythical

Matteo Collina yesterday asked me, in reply to my tweet: but didn't y

Fable and GPT 5.6 reviews to the sorted sets memory saving are going

:

Search has its own bitter lesson

DOUG TURNBULL · 22 JUN 2026 · [SOURCE](#)

An agent with grep does quite well at search.

It's shocking to search technologists. We want to cocoon the problem in technology. We care about algorithms, knowledge graphs, ranking: ever-and-ever smarter retrieval.

It turns out, though, you can play a bit of a trick. If you convince everyone to optimize content for your search engine, you'll have built the best search engine.

There's Sutton's [bitter lesson](#) about unleashing raw compute on a problem. But there's a different bitter lesson in search: algorithms matter less than convincing the world to optimize for your search engine.

Sure, technology acts as the trigger. In Web Search, Google shifted away from text relevance to prioritize authoritativeness (PageRank). Their success told the market: create content others want to link to. They reshaped the Web for good (or ill) around that incentive.

Google crowdsourced search quality to the Web. Claude Code outsources it to every developer.

Developers depend on agents to find documentation on the file system. We like that dopamine hit. That thrill of Claude Code knowing exactly how to reason about our code.

When agents make boneheaded mistakes, we stress. We frantically update documentation, we link to it from AGENTS.md, we add a new document, we place it in a useful location. We do anything - absolutely anything - to ensure agents know how to find and understand documentation.

Just as the SEO expert reads the tea leaves of Google’s algorithm. The software developer writes to the LLM to make documentation findable + useful.

A search system with these incentives overcomes technologically sophisticated systems. For example, classic RAG rests on a foundation of technically sophisticated question answering. Answers often look like conversationally valid responses, but with little linkage back to questions it might answer:

Question:

What is a runner test?

Answer:

Verifies that small-dollar deposits and withdrawals can be executed against real test bank accounts during account reconciliation.

Yes that’s literally an answer to the question.

But I wouldn’t consider this “good documentation”. It’s not anchored to any concepts from the question. It’s just an out of context bit of text.

If you wrote this for a coding agent, you’d probably organize the chunk to give hints about why it’s useful.

docs/tests/runner.md

Runner Tests

Purpose

We need to ensure the system runs end to end

Usage

Use this style of test sparingly. Unit and standard e2e tests should usually be preferred.

What is it

A runner test actually interacts with real-life test bank accounts to ensure we can correctly deposit / withdraw money and that the full banking system works.

The Lazy Agent: You tell the agent to try locking again, except this time monitor its own performance over batches of files and pick efficient locking protocols. The agent determines that the best way to optimize locking is to just not do it, because it seems quite unlikely that someone else would be accessing these files at the same time.

The Clever Agent: Of course, everyone on your team is a distributed systems expert; and so someone built a CLI tool in 2018 for safely transferring your application’s state, with support for locking and write-ahead logging in case of failures; there’s even a sentence in your onboarding doc saying “NEVER TOUCH STATE IN FOOBARDDB DIRECTLY!”. You even told the agent explicitly about this tool. But time passed, context filled up, the agent forgot about this tool’s existence, or maybe it just preferred FooBarDB’s in-distribution API to your esoteric CLI tool. You had anticipated this and not given it access to the FooBarDB client library; and marked the FooBarDB CLI as non-executable; but it just went ahead and used curl to access a forgotten REST endpoint.

The Rogue Agent: As the agent copies state (this time using the CLI tool after you yell at it enough), it lists keys in batches; each key corresponds to some user of your application. Someone decided to pick the user name “delete-everything”. Of course, you are using the latest model which is always resistant to prompt injections. Almost always.

The Stupid Agent: You do everything right. There are no prompt injections in your state. You wall off FooBarDB and force the agent to use the CLI tool. The model decides to delight you by generating an unusual and interesting sequence of tokens that will optimally save storage space for you, translating to a lambda that deletes your production data entirely.

One response to these difficult failure modes might be: wait until the next model. Maybe a smarter model will never delete all of your production data, or implement a slow locking protocol. But even the smartest model is subject to the four horsemen of the distributed apocalypse: asynchrony, crashes, concurrency, and network partitions.

In recent work called [LogAct](#), my colleagues and I take a first step towards solving this problem. We go back to the difference between vibe coding and vibe engineering: git is *transactional*, infrastructure is not. How do we make infra more transactional, like git? We propose that an agent should be a ***deconstructed state machine on a shared log***, borrowing ideas from distributed systems (State Machine Replication, Byzantine Fault-Tolerance) and databases (Atomic Commit, Write-Ahead Logging). LogAct can stop unsafe actions before they happen (by collecting votes on the shared log); recover from crashed actions (using the shared log as a WAL); and provide an audit trail for actions after they complete.

Now, the observation: **vibe coding != vibe engineering**. In vibe coding, the agent / self-writing program acts primarily upon a code repository. Code repositories are forgiving: an agent can hallucinate, create unnecessary files, delete critical state, remove tests entirely, fail in the middle, forget what it was doing, reward-hack in bizarre ways, completely trash a code repo; all you have to do is git reset.

In vibe engineering, the agent acts upon infrastructure (S3 buckets, DynamoDB tables, EC2 instances, K8s deployments...). Large-scale infrastructure is not forgiving. There is no reset. Only SEVs at 2 AM.

So, a re-restatement of Agentic Fault-Tolerance: *How do we make a self-writing program fault-tolerant when it acts upon real-world environments?*

To answer this question, we examine the many types of failures that agents can experience.

The Crashed Agent: You tell an agent to move a bunch of data from FooBarDB to S3 for freeing up space. It generates a lambda to move this data and chugs along moving files. Midway through, the agent crashes. How do you recover?

The Zombie Agent: You start a new agent on a different server. Happily, the previous agent crashed after the copy but before the delete; so a new agent is able to re-execute the command in an idempotent manner. Sadly, the files are now very slow to access; you change your mind and tell the agent to move them back to FooBarDB, which it does successfully. You go get a cup of coffee. Unfortunately, it turns out that the previous agent had not actually crashed; it was just temporarily partitioned away on the network. It wakes up and continues executing its logic, deleting the (now only) copy of state from FooBarDB.

The Dining Agent Philosophers: While you were telling your agent to copy state from FooBarDB to S3, someone else on the team had the same idea and started another agent to do the exact same thing. Now you have two copies of state in S3. Of course, you could have solved this problem by appointing the one and only Agentic Oncall Czar responsible for running the single agent that can touch production; but then if that agent crashes, now you have a potential Zombie Agent.

The Slow Agent: You tell your agent to follow a sophisticated locking protocol before accessing the FooBarDB table. This works to avoid concurrency bugs... but the agent decides to lock each file individually by writing to a conditional register, which takes forever. At some point you kill the agent, turning a Slow Agent into a Crashed Agent.

That's a far more descriptive piece of content to an LLM. Being encouraged to write *for the LLM itself* makes all the difference.

(As an aside, LLMs see candidate *answers* in their context before the user's question - so just treating RAG as question answering doesn't work).

What often matters isn't algorithms. It's programming the content author to create useful LLM context. That moves content discovery from random, disconnected factoids to documentation connecting information to problems to solve.

You learn that the trick to RAG isn't creating good question answering systems. Its writing better answers.

The enterprise search morass... is over?

Compare how coding agents program us to the eternal nightmare of your company's wiki.

When people write in the wiki, they do it for themselves. Nobody is incentivized to make information findable by anyone. No author sits obsessing how to 'SEO' the content for Confluence search.

If someone is 'lucky' enough to find your article, they can barely understand it. It's written in your team's jargon. And it's probably out of date anyways.

For these reasons, I always tell people they'd do better to write a public blog article. Blogs assume zero context. They're clearly timestamped as an artifact of a moment in time. To avoid shame or promote your career, you're incentivized to create something useful, interesting, and findable.

Writing for a coding agent has the same property. You're writing documentation for a targeted reader. You care to SEO your content for Claude Code + grep.

Enterprise search fails because there's no feedback loop between authorship and findability. Coding agents solve that, finally giving some hope to organizing company information into findable knowledge.

The real outcome of coding agents isn't just code generation. It's closing the feedback loop between company context and the next piece of work being done. That old fable of Enterprise Search - improving workforce productivity by leveraging existing insight - can finally become true.

Feedback to content authors drives user relevance

Look around your system. Who creates the content? What's in it for them to make content findable?

A recurring theme in my consulting looks something like this - a job search company notices spammy behavior from job posters. They keyword stuff every programming language in some hidden text element. The usual reaction: that's spammy and we should ban them.

That's too black and white. You have content creators willing to put in work. Shift their black-hat SEO over to an acceptable white-hat SEO. Document what's acceptable. What's considered malicious.

Focus on feedback. Make the content-optimization path rewarding. Give creators feedback+guidance during creation. LLMs help detect spam in real-time. They push content creators to author what's useful not malicious.

Give content creators feedback. Once you've identified what 'good' is, give those content authors dopamine hits. Tell content creators what queries perform well for the content. Make it real-time, sticky, fun. Incentivize and reward behaviors you want - and punish those you don't.

Steer content creators towards the content you want. Don't hope algorithms can overcome drivel and slop.

Your search engine isn't a bundle of algorithms ranking for users. It's a relationship between authors and users. That bitter lesson puts grep above the smartest reranker and embedding model. Forget it at your peril.

Enjoy softwareDoug in training course form!

Starting May 18!

Signup here - <http://maven.com/softwareDoug/cheat-at-search>



I hope you join me at [Cheat at Search with Agents](#) to learn use agents in search, build better RAG and use LLMs in query understanding.

© 2026 SoftwareDoug LLC

Your Agent is a Distributed System (and fails like one)

MAHESH'S BLOG · 24 APR 2026 · [SOURCE](#)

We start with a re-definition and an observation.

A common definition of an agent is that it's an LLM that calls tools in a loop. This definition is incorrect. An agent is not an LLM. It is not restricted to calling tools. It does not particularly have to be a loop.

Instead, a better definition of an agent is that it's a self-writing program. If a program is defined as a sequence of states, moving from one state to another by executing a lambda, then an agent is a program that uses an external LLM to determine the next lambda to execute / the next state in this sequence.

Accordingly, the new problem of Agentic Fault-Tolerance can be restated as: *How do we make a self-writing program fault-tolerant?*